DELTA STRATEGY CO.

A Risk-Centric Portfolio Architecture for Convex Returns

Public Non-Confidential Executive Abstract

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1. CONFIDENTIALITY & SCOPE

This abstract is a public, non-confidential overview prepared for professional reference. It discloses no proprietary signals, formulas, code, model parameters, asset universe, thresholds, or implementation details. It documents the research program's structure, lineage, and conceptual approach only. Nothing herein constitutes an offer, a performance representation, or a solicitation of investment, and detailed methodology remains private.

The program described here was early-stage and exploratory. This document summarizes its themes at the level of general direction, not mechanism, and should not be read as describing any specific model or algorithm — nor any research conducted outside the program.

2. OVERVIEW

Systematic strategies degrade out of sample for well-understood reasons: non-stationarity, regime change, overfitting, and signal decay. Delta Strategy was a pre-AUM systematic research program built to confront these failure modes directly — by treating uncertainty and downside consequence as the primary design constraint rather than as residual noise.

The framework was organized around a risk-first premise: exposure is earned, not assumed. The system held a conservative, low-exposure baseline and scaled position size only as probabilistic evidence accumulated that a thesis was robust to the prevailing environment and to stress. Sizing followed a Kelly-style capital-growth discipline, deliberately tempered for capital preservation — built to survive and compound under uncertainty rather than tuned to maximize apparent edge. Shifts in macro regime — policy, liquidity, and cross-asset behavior — and finer-grained changes in volatility and correlation structure were treated as first-class inputs that gate exposure, rather than as noise.

A second priority was parsimony. Rather than expanding data pipelines and feature counts, the framework concentrated on a small set of high-conviction, robust features and committed exposure only where asymmetry was supported by evidence. The intent was institutional-grade risk discipline without the operational fragility and overhead of heavier systems — precision through lean probabilistic structure rather than scale or speed.

3. ORIGIN & RESEARCH LINEAGE

Delta Strategy Co. operated (2023–2025) as a small systematic research group under senior buy-side (former Guggenheim) leadership and review, with a mandate to carry a risk-first portfolio architecture toward a deployment-aware standard rather than an in-sample one.

The framework consolidated a research line developed across three settings prior to and alongside the program: Bayesian and stochastic inference methods — hierarchical modeling, MCMC, Itô/Langevin dynamics, and regime and state-space models — originating in academic time-series research, including computational neuroscience; position-sizing and cost discipline proven in live, independent systematic trading under real capital and real transaction costs; and risk-governance architecture — cross-asset VaR/CVaR measurement, drawdown governance, and scenario shock propagation — drawn from practitioner advisory work for investment and family-office mandates.

At Delta, these strands were drawn together into an early-stage, modular prototype — exploratory rather than final — combining probabilistic, regime-aware assessment with preservation-first position sizing, supported by schema-controlled research infrastructure and a decision layer translating market state into exposure caps and drawdown thresholds.

The program is best understood as a formative stage: a working origin for ideas that individual members have since refined, in substantially more rigorous form, in independent settings. Delta was pre-AUM; its results derive from internal validation under deployment-aware standards, not from managed outside capital, and the program makes no performance representation.

4. DESIGN PILLARS

The following themes describe the prototype's directional logic and structural posture. They are conceptual only and disclose no implementation specifics.

- Bayesian, probabilistic inference, with regime awareness treated as a first-class input to exposure decisions
- Growth-versus-ruin capital deployment in the Kelly tradition, tempered for survival under uncertainty
- Constraint-first capital preservation: exposure caps and drawdown thresholds translating market state into position discipline
- Structural gating to limit drawdowns during volatility expansion
- Signal sparsity: a deliberately small set of high-conviction, robust features

All mathematical specifications, execution details, and proprietary signals are withheld.

5. VALIDATION STANDARDS

The prototype was evaluated under standards intended to mirror what an institutional risk committee would vet before allocating:

- **Out-of-sample integrity.** Research parameters were fixed in advance of evaluation and walk-forward testing was run genuinely out of sample, without data snooping.
- **Cost realism.** A priori transaction-cost and slippage modeling charged on turnover, mapping where edge degrades under realistic frictions.
- **Data integrity.** Strict point-in-time, no-look-ahead data handling.
- **Stress.** Monte Carlo, jump-diffusion, and synthetic-data simulation across multi-cycle regimes, including the Dot-Com, Global Financial Crisis, and Euro-crisis periods, with explicit drawdown governance and capital-preservation review.
- **Reproducibility.** Schema-controlled research infrastructure with audit trails and reproducible retrieval.

These standards describe process, not results. Validation was internal and exploratory, and no performance representation is made.

6. RESEARCH & ATTRIBUTION

This abstract documents the structure, lineage, and validation standards of a pre-AUM systematic research program developed at Delta Strategy Co.

Senior review & program oversight — Jeff Traister (formerly Guggenheim Partners)

Scientific advisor — Dr. Katsushi Arisaka (UCLA, Distinguished Professor)

Lead researcher & architecture — Syed B. Hydari

Coordination & support — Three members

Successor research pursued independently by Syed B. Hydari in academic and professional settings is separate from this program. This document neither describes such work nor makes any claim over it.

The complete white paper is held in private circulation and is available to qualified professional parties on request. © Delta Strategy Co. All rights reserved.